RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

Bachelor of Science in Applied Sciences
Third Year - Semester II Examination – March - May 2026

COM 3401 – DATA COMMUNICATION AND NETWORKING

Time: Three (03) hours

- This paper contains **Five (05)** questions in **Three (03)** pages.
- Answer **ALL** questions.

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1. a) Draw a suitable block diagram of a simple data communication system, clearly showing its key elements.

(02 marks)

b) Using the diagram drawn in part a), explain how data is transmitted through the system.

(04 marks)

c) Explain the following tasks in a data communication system:

i. Flow control.

ii. Recovery.

(04 marks)

d) What is data transmission? Explain its importance in data communication systems.

(04 marks)

e) Define the following terms in data transmission:

i. Direct link.

ii. Simplex mode.

iii. Duplex mode.

(06 marks)

2. a) What is the effective bandwidth of a signal, and why is it important in data transmission?

(04 marks)

b) Briefly explain how analog data is transmitted using digital signals.

(04 marks)

c) Explain the following modulation techniques:

i. AM.

ii. FSK.

iii. Differential PSK.

(06 marks)

d) Briefly explain the three (03) major types of transmission impairments.

(06 marks)

3. a) List and briefly state five (05) important uses of computer networks in scientific applications.

(05 marks)

b) Explain the use of the following devices in computer networks:

i. Modem.

ii. Gateway.

(04 marks)

c) Explain what a network topology is using the star topology as an example.

(05 marks)

d) Briefly explain the importance of the following in the context of layered network architecture:

i. Having clear interfaces between layers.

ii. Error handling in the OSI Data Link layer.

(06 marks)

4. a) Define framing in data communication and explain why it is needed.

(04 marks)

b) Explain the character count framing method.

(04 marks)

c) Describe the multiplexing technique widely used in radio and television broadcasting.

(04 marks)

d) Why is statistical TDM more efficient than synchronous TDM?

(04 marks)

e) Five digital channels, each with a data rate of 64 kbps, need to be transmitted over a single link using synchronous Time Division Multiplexing (TDM).

i. Calculate the minimum required data rate of the link using synchronous TDM.

ii. Compare the calculated data rate in part (a) with the minimum required data rate when statistical TDM is used and explain the difference.

(02 x 2 marks)

5. a) State four (04) specific functions performed by transport layer error control protocols.

(04 marks)

b) State the main difference between the services provided by connection-oriented and connectionless transport protocols.

(04 marks)

c) What is classful addressing in IPv4?

(02 marks)

d) Explain how to determine the class of an IP address using its binary notation. For simplicity, ignore the class reserved for future use.

(04 marks)

e) Consider the following IP address and network mask given in binary notation from a classless block of addresses:

IP Address: 00010100000100000000010000010010

Network Mask: 11111111111111111111111111110000

i. Write the IP address in dotted decimal notation.

ii. Write the network mask in CIDR notation.

iii. Find the number of addresses in the block.

iv. Determine the first and last addresses in the block.

(06 marks)

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